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Psychological Safety and Hierarchy in Operating Room Debriefing: Reflexive Thematic Analysis



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ABSTRACT

Introduction: Debriefing is a team discussion in a constructive, supportive environment. Barriers exist to consistent, effective team debriefing in the clinical setting, especially in operating theaters. The purpose of this study was to gain insights from frontline workers on how to set up an effective debriefing policy for our operating room.

Methods: This was a qualitative study in which we interviewed operating room workers in a tertiary children's hospital. Interviews were audio-recorded, transcribed, and coded. Data were analysed using the reflexive thematic analysis technique within a critical realism paradigm.

Results: Interviews were analysed from 40 operating room staff: 14 nurses, seven anesthetic technicians, seven anaesthetists, and 12 surgeons; 25 (65%) were female. The three key themes were (1) "commitment to learning"—healthcare workers are committed to teamwork and quality improvement; (2) "it is a safe space"—psychological safety is a prerequisite for, and is enhanced by, debriefing; and (3) "natural leader"—the value of leadership, but also constructs around leadership that maintain hierarchies.

Conclusions: Psychological safety is both a prerequisite for and a product of debriefing. Leadership, if viewed as a collective responsibility, could help break down power structures. Given the results of this study and evidence in the literature, it is likely that routine debriefing, if well done, will improve psychological safety, facilitate team learning, reduce errors, and improve patient safety.

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Introduction

Surgery is a high-risk undertaking in which professionals from a variety of specialties work together on a time-critical operation in which safety is of the utmost importance. Similar to teams in aviation, motor racing, and sport, surgical

teams can be described as action teams. Where surgical teams differ is the diversity of roles within the group: nurses, anesthesiologists, technicians, and surgeons, from students to seniors. To perform effectively, team members must feel safe about sharing observations and opinions with the rest of the team;¹ however, diverse surgical teams are often beset by intergroup

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rivalries based on profession, gender, and seniority divides. The resulting power dichotomies inhibit effective teamwork.

One way to counter the problem of hierarchy in surgery may be through regular team debriefing. Debriefing is a process that allows individuals to discuss team performance in a constructive and supportive environment.² Debriefing has been linked to improved performance, teamwork, and communication, and can assist with error identification.³

Debriefing is one component of the World Health Organization Surgical Safety Checklist (SSC). Initially a three-point checklist involving a sign-in, time-out, and sign-out,⁴ the SSC was expanded to include five steps by adding a briefing discussion at the beginning of the operating list and debriefing at the end of the list.⁵ The sign-in, time-out, and sign-out are now performed with a high level of compliance in operating rooms throughout the world.⁶ In our tertiary pediatric hospital, compliance with the SSC is 100% but compliance with debriefing (which is not mandated at the present time) is <10% (personal communication, Starship Children's Hospital Operating Rooms, 2022). Debriefing in surgery is challenging to implement⁷ possibly because it involves a coordinated team discussion rather than a simple checklist, or for reasons related to power dynamics. Given that debriefing is central to other fields including medical simulation,^{2,8} the aviation industry,⁹ the space industry,¹⁰ and the military,¹¹ surgery appears to be missing out on the benefits of debriefing in clinical practice.

Our operating theater introduced debriefing in 2017. Despite positive feedback, debriefing did not become routine practice throughout our operating rooms, being performed in only a minority of operating lists. Given the potential value of debriefing for surgical teams but the challenges of putting it into practice, we were interested in the views of our operating theater staff as to how, why, when, and where to debrief. Therefore, the purpose of this study was to gain insights from frontline workers on how to set up an effective debriefing policy for our operating room suite, to explore operating room workers' experiences with debriefing, and to critically reflect on the meaning of debriefing to psychological safety, hierarchy, and teamwork.

Materials and Methods

This study was registered with the institutional research office and received ethical approval from the Auckland Health Research Ethics Committee (#3228).

The setting was the operating room of a children's hospital. A simple debriefing structure was developed which consisted of three phases: (1) introductions and ground rules (confidentiality, respect, share the air-time); (2) discussion (what went well, what could have gone better); and (3) take-home actions. This structure was used mostly in pediatric surgery and in ear, nose, and throat operating lists. Pediatric cardiac surgery debriefed all operations using an extended checklist that included perfusion time, blood loss, and other audit data. Apart from cardiac surgery, debriefings were performed inconsistently depending on the surgeon's preference for debriefing and were not part of the operating room culture.

The inclusion criteria were operating room staff including nurses, anesthetic technicians, anesthetists, and surgeons, who worked regularly in the children's operating suite. Excluded were students and other temporary visitors to the operating rooms. The sampling strategy was purposive, with care taken to interview a range of professions and specialty groups. Sample size was determined pragmatically by the number of interviews able to be performed over the study period, December 2021 to February 2022, but also guided reflexively as recommended by Braun and Clarke,^{12,13} and Malterud's "information power" concept helped guide the decision as to when to stop recruiting.¹⁴

The semistructured interview guide was developed by the researchers to cover the items of interest based on our experience with surgical debriefing (available in [Appendix S1](#)). Limited testing of the interview guide was undertaken by mock interviews between three researchers. Two researchers (C.M. and E.S.) performed all interviews. They received interview training from an independent cultural expert, from the senior author, and by practicing mock interviews. Interviews began with making personal connections and explaining the study, and then proceeded with open-ended questions with an emphasis on letting the interviewee tell their story and do most of the talking.

Operating room staff were made aware of the project through presentations at teaching and business meetings. Two researchers (C.M. and E.S.) actively invited participants to take part in an interview. Participants gave written informed consent. Interviews were conducted between December 2021 and February 2022. All interviews took place in person in a private space. Interviews were audio-recorded using a mobile device and transcribed verbatim. Participants received a copy of their transcript for editing and approval. Transcripts were de-identified and labeled with a random number generated from the random sequence generator service of random.org.¹⁵

The qualitative methodology used for this research was reflexive thematic analysis.¹³ The theoretical framework was one of critical realism.¹⁶ We followed the process of reflexive thematic analysis described by Braun and Clarke.¹⁷ Briefly, coding and theme development were recursive processes involving immersion in data, review of relevant literature, and deep reflection. Immersion in the dataset was by transcribing (performed by C.M. and E.S.), reading, and re-reading interview transcripts. Two researchers (C.M. and J.H.) coded the data separately and met regularly to discuss and mold themes. Quality and reporting were guided by Braun and Clarke¹⁸ and Elliot et al¹⁹ and used the following checklists: Braun and Clarke's 15-point checklist for good thematic analysis²⁰ as shown in [Table S1](#); Nelson's conceptual depth scale²¹ as shown in [Table S2](#); Braun and Clarke's 20 questions for evaluating thematic analysis manuscripts for publication¹⁸ as shown in [Table S3](#); and the Standards for Reporting Qualitative Research²² as shown in [Table S4](#).

Results

Of 41 interviews conducted, one recording failed leaving 40 interviews to be transcribed and analysed. Interviewees were predominantly female ($n = 26$, 65%). Interviewee

characteristics are shown in Table 1. Each interview lasted approximately 10-20 min. Maps of initial and refined themes are presented in Figures S1 and S2. We developed three main themes relevant to psychological safety and hierarchy: “committed to learning”, “it is a safe space”, and “natural leader”. Additional quotations for each theme are presented in Table 2.

Committed to learning

A strong theme was how committed frontline workers are to learning and quality improvement. Learning was seen as one of the most important things we do in healthcare and debriefs were seen as a vehicle for learning.

“Is there anything that we can learn from this, you know that’s almost the most important thing in modern medicine and healthcare, isn’t it? You know it’s, ‘what can we do better next time so that we can avoid this situation?’” (Participant 4)

It was striking that staff are not just committed to individual and team learning; they want to see organizational change too. They wanted debriefs to be incorporated into a quality improvement process so that issues raised were dealt with at an organizational level. If debriefs did not lead to quality improvement, then they were seen as a waste of time.

“It’s kind of like you just get stuck in Groundhog [Day] with the same – you’re not coming to a conclusion or fixing the problem...” (Participant 26)

Conversely, if debriefs lead to learning, they are seen as immensely valuable.

“I’ve found it absolutely one of the most valuable things that we’ve had instituted here.” (Participant 38)

It is a safe space

The importance of psychological safety in the team was a strong theme throughout the interviews. Psychological safety was promoted by debriefing and was a prerequisite for effective debriefing.

“And then also just making sure that if someone does say something they’re not going to be shot down or they’re not going to be – yeah, just making sure it’s a safe space to do so.” (Participant 11)

Interviewees felt that consistent debriefing would facilitate better communication, help flatten the hierarchy, and create a more inclusive work environment. Many participants recognized the importance of acknowledging positives during debriefs. Positive feedback is an effective way to encourage speaking up. Developing a structure that makes space for positives (“what went well”) was helpful. Several respondents commented that debriefing is a good way to unpack what had happened during the day, unwind a bit, and to go home

leaving work at work, which would clearly carry co-benefits for mental health and resilience.

“I think the debriefing actually finishes off the day and creates a good work environment and lets people kind of put work aside.” (Participant 36)

Operating room staff needed to feel they were in a safe place to debrief effectively. Safety took various forms: assurance of confidentiality, a nonjudgmental and respectful environment, and a flat hierarchy. People need to feel safe from being judged. Debriefs need to be conducted in a climate of respect. Creating an environment in which everyone, including inexperienced staff and students, can feel free to speak up is a challenge when there is an obvious hierarchy arrangement. There was an interplay between the tone set in the preoperative briefing and that of the postoperative debriefing and an interplay with the general culture of the operating room environment.

“I think that starts from before the brief. So, I think having everyone there from the beginning, having an expectation that there’s a flattened hierarchy, we would welcome people speaking up, and that we use individual names, we introduce each other at the brief, we highlight how important each team member is.” (Participant 36)

Natural leader

The idea that there would be a “natural” leader was a theme among participants, although opinions on who or what that leader should be varied.

“I think that’s a difficult one, actually. Because it could be different people on different days... it shouldn’t really be a set person every time. Because sometimes the surgeons change, sometimes the nurse in charge will pop out and then come back at the end. So really, there should be a

Table 1 – Interviewee characteristics.

Characteristic	n = 40
Gender	
Female	25
Male	15
Profession	
Nursing	14
Anesthesia	7
Anesthetic technician	7
Surgeon	12
Surgical specialty	
Pediatric surgery	7
Orthopedic surgery	2
Cardiac surgery	2
ENT	1

Table 2 – Additional quotations by theme and subtheme.

Theme/subtheme	Quotation	Participant #
Committed to learning		
Team learning	It is just a way to kind of come together with the team at the end of the day and talk about any issues and try and think of possible solutions...	10
	Sometimes there are quite good key learning points and from theater, well if every theater did a decent debrief at the end of the day, I think that there might be things that might come up consistently, repeatedly, that actually we can learn from—learning debriefs...	36
	Is there anything that we can learn from this, you know that is almost the most important thing in modern medicine and healthcare, is not it? You know it is, “what can we do better next time so that we can avoid this situation?”	4
Organizational learning	It is kind of like you just get stuck in Groundhog [day] with the same—you are not coming to a conclusion or fixing the problem...	26
	... although if there is any big issue, I think if there is anything that needs improving, it could be good to record... otherwise it would not get fixed.	13
	Can you close that? Can you come back to us in 2 wk to know how you progress?	14
	I have found it absolutely one of the most valuable things that we have had instituted here.	38
It is a safe place		
Debriefing promotes psychological safety	By doing briefing and debriefing it actually helps to break down barriers between different subgroups.	2
	So sometime, often when the list is run really well, I think it is really important to acknowledge that too, everybody to kind of be grateful to each other.	36
	Everybody responds very differently to trauma and to, you know, critical incidents, and sometimes people just need that little bit of help and support to sort of express how they feel. Because otherwise you may end up losing these people because they just do not want to do this job anymore.	4
Psychological safety facilitates debriefing	I think the debriefing actually finishes off the day and creates a good work environment and lets people kind of put work aside.	36
	I think that starts from before the brief. So, I think having everyone there from the beginning, having an expectation that there is a flattened hierarchy, we would welcome people speaking up, and that we use individual names, we introduce each other at the brief, we highlight how important each team member is.	36
	In terms of like if people want to—anyone can say something, like it is not a judgmental environment.	26
	I think also that needs to be emphasized that it is confidential	38
	I guess sometimes that is where there might be issues with people not wanting to speak up if they feel that there is a hierarchy.	22
	It is a longer-term thing... I think there is an old-school way of practising that you might not have even experienced that I experienced as a junior doctor—it is fading out where there is quite a lot of barriers to communication with senior people. And I think that is improving, definitely, but I think it is something that we generate every day through our interactions with other staff members.	28
One natural born leader		
Surgical leadership	In my experience, I think it is best to have a surgeon or anesthetist be onboard and leading it. I feel like they are more of like a leader in the room and... in my experience, people appear a lot more engaged when it is an anesthetist or a surgeon or someone quite senior in the room to be leading it.	31
Nursing leadership	The experience that generally I have is that the nurse that is coordinating the list for the day runs it, and I think that allows for a lot more of an inclusive approach.	20
	I think that is a difficult one, actually. Because it could be different people on different days... it should not really be a set person every time. Because sometimes the surgeons change, sometimes the nurse in charge will pop out and then come back at the end. So really, there should be a natural leader that emerges, and sort of leads the debrief. But probably the person that needs to instigate it should be the nurse in charge, because otherwise surgeons might just forget to do it. So, it should be instigated by the nurse in charge, but then the natural leader should just do the chatting.	18
It is about skills	I do not think it should be any particular person, I do not think. But I think the person needs to be able to lead it, so they need a sense of confidence, and they need to be able to understand when the most appropriate time is for debriefing.	7

natural leader that emerges, and sort of leads the debrief. But probably the person that needs to instigate it should be the nurse in charge, because otherwise surgeons might just forget to do it. So, it should be instigated by the nurse in charge, but then the natural leader should just do the chatting.” (Participant 18)

Many interviewees viewed senior doctors, the surgeon or the anesthetist, as the main leaders in the operating room.

“In my experience, I think it’s best to have a surgeon or anaesthetist be onboard and leading it. I feel like they’re more of like a leader in the room and... in my experience people appear a lot more engaged when it’s an anaesthetist or a surgeon or someone quite senior in the room to be leading it.” (Participant 31)

Other interviewees felt that nurses were best suited to provide a comprehensive overview of operating room events and tended to follow a more structured and systematic approach.

“The experience that generally I have is that the nurse that’s coordinating the list for the day runs it, and I think that allows for a lot more of an inclusive approach.” (Participant 20)

Still, other interviewees felt that anyone within the multidisciplinary operative team should be able to lead an operative team debrief. Responses suggest that an individual’s experience and communication skills may be more relevant to a leadership position than their title or specialty. Therefore, the leadership role can be flexible and negotiated.

“I don’t think it should be any particular person, I don’t think. But I think the person needs to be able to lead it, so they need a sense of confidence, and they need to be able to understand when the most appropriate time is for debriefing.” (Participant 7)

Discussion

By listening to frontline workers, we have attempted to understand what is important when working within an operating room environment. The themes reflected a rich sense of the value of debriefing for learning, culture, and togetherness as a team. Debriefing’s influence on a culture of gratitude and psychological safety came through. Our research has established a foundation for various factors that relate to psychological safety including power hierarchies and perceptions of leadership.

Committed to learning

The commitment to quality, safety, and organizational learning by frontline staff is not always appreciated and organizations can miss opportunities to tap into the intellectual value of employees. In a qualitative study of frontline healthcare workers in London, Lalani et al²³ found that “participants from across health and social care expressed their ambition to

work collaboratively and maximize opportunities for formal and informal learning.” Fathy Ahmed et al²⁴ demonstrated a strong positive correlation between goal orientation in staff nurses (commitment to learning and performance) and organizational learning culture (dialogue, team learning, collaboration, empowerment) and recommended that nurses are empowered to participate in decision-making, problem-solving, and in achieving organizational goals. Our research showed that people working in theater are committed to learning and improvement. Debriefs were seen as a vehicle for individual and team learning, with the potential to contribute to organizational change. Overall, committed staff in healthcare are eager to learn and improve, and organizations would do well to value their input and incorporate their suggestions into quality improvement processes.

Psychological safety

Our study uncovered a prominent theme concerning psychological safety in the theater environment, which was viewed by participants as paramount for effective debriefing. According to Edmondson, psychological safety ensures that individuals are not punished or humiliated for expressing ideas, questions, concerns, or mistakes.²⁵ Therefore, psychological safety is shaped by broader cultural norms within the organization. Teams who welcome feedback have the healthiest work environment by supporting staff that report problems without fear of retribution.^{26,27}

To create trust, some institutions incorporate programs that acknowledge or give positive recognition for reporting.²⁸ Eliminating the fear of consequence is crucial to building trust and establishing that reporting will not negatively affect the staff member voicing their concerns or the member of staff involved in a report.²⁹ Elimination of fear of consequence builds on staff trust, ensures positive feedback, and provides learning opportunities from the reported event.^{29,30}

Our research recognized the importance of acknowledging positives during debriefs. Positive feedback was seen as an effective way to encourage individuals to speak up and develop a structure that makes space for positives. Developing a structure that makes space for positive feedback was seen as an effective way to encourage individuals to speak up.

Hierarchies are well-recognized in clinical healthcare and can impede safety by contributing to a reluctance to speak up.^{31,32} Hierarchies can obstruct interprofessional teamwork and collaboration leading to a hostile work environment that impacts individual and team psychological safety.^{25,33} However, hierarchies can also provide a template for expectations of interaction that generate and sustain the structure and stability necessary to enable the formation and efficiency of the team.^{32,34,35} To promote a just culture and psychological safety, it is crucial to recognize and address the negative effects of power hierarchies in the workplace. By doing so, healthcare professionals can foster an environment that values open communication, teamwork, and safety.³²

Leadership

Our research showed that participants had varying perspectives on leadership in the context of medical debriefing. The

notion of leadership is subjective and lacks consensus. The idea of a “natural leader” can reinforce hierarchical constructs and encourage dominant individuals to take over, which can hinder team performance and psychological safety.^{36,37} Although limited research exists on what constitutes a leader in medical debriefing, our study sheds light on the importance of addressing leadership and hierarchy in promoting effective teamwork and creating a culture of safety. Research suggests that those who are perceived to be leaders due to their hierarchical standing may not necessarily be capable of leading teams due to insufficient training and are therefore no more capable than anyone else to lead debriefing.³⁸

An alternative approach is collective leadership in which roles and responsibilities are shared, distributed, or rotated among team members.^{39,40} By distributing leadership across teams, formal and informal leaders work together to generate actions, promoting involvement in decision-making and a sense of belonging within a group.^{39,40} Flat hierarchies are gaining popularity in healthcare, as they provide the flexibility and equality needed in a caring environment where all individuals should feel safe to raise concerns and voice opinions.⁴¹ With flat hierarchies, employees have more responsibility for each staff member, as there are more people available for support and guidance.⁴¹ Despite various leadership approaches introduced to healthcare, there is no “one size fits all.”

Resonance

Our findings will resonate with others who practice debriefing in the clinical setting. One of the “big tent” criteria for excellent qualitative research described by Tracy (2010) was resonance, which included transferability and naturalistic generalization.⁴² In our practice, we have started to transfer the findings of this study by (1) nominating a nurse educator as debriefing coordinator to work with quality and safety on issues raised in debriefs and to feed back to staff; (2) working with our simulation service on a debriefing short-course for theater staff including nurses, anesthesiologists, and surgeons; and (3) designation of the nurse team leader of each operating list as the person responsible for initiating and leading the debrief at the end of the list. We hope others will find naturalistic generalization of the themes in this study in their practice.

Limitations

Debriefing had been performed on a relatively ad hoc basis in our operating rooms, so interviewees had varying degrees of experience. Further research on the experience of operating room workers with regular debriefing would be of value. The study was set in a tertiary pediatric hospital. Subsequent research across a range of hospitals and specialties could provide further insights. Voluntary participation in this study could introduce sampling bias. Although the results of this research were shared with the theater unit where the research took place, formal reviewing of the themes with interviewees (member-checking) was not performed. In keeping with the critical realist paradigm, some of the themes were more descriptive while others were developed (in the discussion)

along a more constructionist (critical) line; however, we believe this will enable readers to reflect on some of the practical issues as well as the social issues around debriefing.

Conclusion

In conclusion, this study of the insights and experiences of frontline workers in the operating room shows that regular, structured debriefs could improve psychological safety, help counter the negative effects of hierarchies and power structures, and promote teamwork. To set up an effective debriefing program, it is essential to create a psychologically safe working environment where individuals can be open and confident in group situations without fear of retribution. By doing so, healthcare professionals can foster an environment that values open communication, teamwork, and safety, ultimately leading to improved patient care and safety.

Author Contributions

C.M. contributed to the acquisition and interpretation of data and drafting the article. E.S. contributed to the acquisition of data and review of the article. M.M. contributed to the interpretation of data and drafting the article. N.M. contributed to the conception of the study and review of the article. L.H. contributed to the conception of the study and review of the article. J.W. contributed to the design of the study and critical review of the article. J.H. contributed to the conception and design of the study, interpretation of data, and drafting the article. All authors contributed to the final approval of the article.

Supplementary Materials

Supplementary data related to this article can be found at <https://doi.org/10.1016/j.jss.2023.11.054>.

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